

Municipal Broadband: The Cost Data

Cities should not build or run broadband networks. The finances do not work. Two studies show why.

Vasquez-Turner surveyed 63 municipal systems built between 2007 and 2021 (402). Median build cost was \$3,140 per premises passed (403). Median bond term was 24 years (403). Median take rate at year five was 27 percent (404). The break-even take rate in her model was 39 percent (404). Only 8 of the 63 systems reached 39 percent by year seven (405). Twelve systems drew on the general fund every year after launch (406). Median annual draw was \$1.9 million (406). Four systems sold the network before year ten (407). Median sale price was 31 cents on the dollar of construction cost (407).

Ng covers the operating side. Ng studied 14 city systems over six years (17). Electronics hit end of life in about seven years (20). A full electronics refresh runs 22 percent of original build cost (20). Nine of the 14 cities carried no reserve for that refresh (21). Median staff size was 11 employees (21). Median technician turnover was 29 percent per year (22). Six cities raised subscriber rates within 30 months of launch (23). Average increase was \$9 per month (23).

Calder Springs is the clearest case. Population 41,000 (Ng 24). Bond of \$58 million issued in 2013 (24). Take rate of 22 percent in 2019 (24). General fund transfers totaled \$14.6 million through 2021 (25). The city cut 6 parks positions in 2020 (25). The property tax rate rose 4 percent in 2021 (26).

Objection. Private carriers skip low-density blocks. Ng found 7 percent of addresses across the 14 cities had one wired option or none (17). The gap is measurable. A subsidy closes it cheaper. Vasquez-Turner priced a voucher program at \$340 per household per year (408). A city build costs \$3,140 per passing up front, plus 24 years of interest (403). The voucher reaches the 7 percent. The build bills the other 93 percent for it.

Recommendation. No city-owned network. Fund vouchers instead. Audit the numbers every 24 months.